

AIRPORT DATA ANALYSIS

```
use flight_analysis;  
select * from airport_project_data;
```

1. Analyze total passenger traffic per route and over time.

```
select  
year, QUARTER, ORIGIN_CITY_NAME, DEST_CITY_NAME, sum(PASSENGERS) as  
total_passengers  
from airport_project_data  
group by 1,2,3,4  
order by year, QUARTER;
```

	year	QUARTER	ORIGIN_CITY_NAME	DEST_CITY_NAME	total_passengers
▶	2025	1	Aberdeen, SD	Minneapolis, MN	1899
	2025	1	Aberdeen, SD	Sioux Falls, SD	0
	2025	1	Abilene, TX	Alexandria, LA	67
	2025	1	Abilene, TX	Chicago, IL	0
	2025	1	Abilene, TX	Dallas/Fort Worth, TX	7068
	2025	1	Abilene, TX	Lubbock, TX	0
	2025	1	Abilene, TX	Phoenix, AZ	0
	2025	1	Adak Island, AK	Anchorage, AK	164
	2025	1	Aguadilla, PR	Fort Lauderdale, FL	2889

#2. Top 5 routes which are the busiest routes having huge passenger traffic.

```
select  
year, ORIGIN_CITY_NAME, DEST_CITY_NAME, SUM(PASSENGERS) as total_passengers  
from airport_project_data  
group by 1,2,3  
order by total_passengers desc  
limit 5;
```

	year	ORIGIN_CITY_NAME	DEST_CITY_NAME	total_passengers
▶	2025	Miami, FL	New York, NY	156277
	2025	New York, NY	Miami, FL	149209
	2025	Orlando, FL	New York, NY	130145
	2025	New York, NY	Orlando, FL	125937
	2025	New York, NY	Chicago, IL	121251

#3. Least busiest routes.

```
select  
year, ORIGIN_CITY_NAME, DEST_CITY_NAME, SUM(PASSENGERS) as total_passengers  
from airport_project_data  
group by 1,2,3  
having sum(PASSENGERS)>0
```

order by total_passengers
limit 5;

	year	ORIGIN_CITY_NAME	DEST_CITY_NAME	total_passengers
▶	2025	Los Angeles, CA	Los Angeles, CA	1
	2025	Miami, FL	Miami, FL	1
	2025	Smyrna, TN	Nashville, TN	1
	2025	St. Mary's, AK	Marshall, AK	1
	2025	Scammon Bay, AK	Sheldon Point, AK	1

#4. Determine average passengers per flight for various routes and airport.

for routes.

```
select  
year, ORIGIN_CITY_NAME, DEST_CITY_NAME, avg(PASSENGERS) as avg_passengers  
from airport_project_data  
group by 1,2,3  
order by avg_passengers desc;
```

for airport. (outhoing)

```
select  
ORIGIN_AIRPORT_ID, avg (PASSENGERS) as avg_passengers  
from airport_project_data  
group by 1  
order by avg_passengers desc;
```

(incoming)

```
select  
DEST_AIRPORT_ID, avg(PASSENGERS) as avg_passengers  
from airport_project_data  
group by 1  
order by avg_passengers desc;
```

```
WITH outgoing_passengers AS (  
    SELECT  
        ORIGIN_AIRPORT_ID,  
        AVG(PASSENGERS) AS avg_passengers  
    FROM airport_project_data  
    GROUP BY ORIGIN_AIRPORT_ID  
) ,
```

```
incoming_passengers AS (  
    SELECT  
        DEST_AIRPORT_ID,  
        AVG(PASSENGERS) AS avg_passengers  
    FROM airport_project_data
```

```

GROUP BY DEST_AIRPORT_ID
),

all_airport AS (
  SELECT
    DISTINCT ORIGIN_AIRPORT_ID AS airport_id
  FROM airport_project_data

  UNION

  SELECT DISTINCT DEST_AIRPORT_ID AS airport_id
  FROM airport_project_data
)

SELECT
  aa.airport_id,
  COALESCE(op.avg_passengers,0) + COALESCE(ip.avg_passengers,0)
  AS total_passengers_travelling
FROM all_airport aa
LEFT JOIN outgoing_passengers op
  ON aa.airport_id = op.ORIGIN_AIRPORT_ID
LEFT JOIN incoming_passengers ip
  ON aa.airport_id = ip.DEST_AIRPORT_ID
ORDER BY total_passengers_travelling DESC;

```

year	ORIGIN_CITY_NAME	DEST_CITY_NAME	avg_passengers
2025	Salt Lake City, UT	Atlanta, GA	40632
2025	Charlotte, NC	Raleigh/Durham, NC	38624
2025	Orlando, FL	Dallas, TX	30432
2025	Dallas/Fort Worth, TX	Santa Ana, CA	29700
2025	Santa Ana, CA	Dallas/Fort Worth, TX	29689
2025	Phoenix, AZ	Philadelphia, PA	28739
2025	Dallas, TX	Orlando, FL	28049
2025	Charleston, SC	Atlanta, GA	27705
2025	Dallas/Fort Worth, TX	Tucson, AZ	27537

#5. Assess flight frequency and identify high traffic corridors.

```

select
  ORIGIN_CITY_NAME, DEST_CITY_NAME, count(AIRLINE_ID) as total_flights
from airport_project_data
group by 1,2
order by total_flights desc
limit 10;

```

ORIGIN_CITY_NAME	DEST_CITY_NAME	total_flights
Seattle, WA	Los Angeles, CA	16
Dallas/Fort Worth, TX	Houston, TX	15
Los Angeles, CA	Seattle, WA	14
New York, NY	Chicago, IL	14
Columbus, OH	Atlanta, GA	13
Chicago, IL	St. Louis, MO	13
Chicago, IL	New York, NY	13
Houston, TX	Dallas/Fort Worth, TX	13
Chicago, IL	Cincinnati, OH	13

#6. Evaluate available seat capacity to understand seat utilization for each airline.

with seating_capacity AS

```
(select
AIRLINE_ID, max(PASSENGERS) as seat_capacity
from airport_project_data
group by AIRLINE_ID
having max(passengers)>0),
```

seat_utilization as

```
(select
ad.AIRLINE_ID, ad.PASSENGERS*100.00/sc.seat_capacity as seat_utilization
from airport_project_data ad join seating_capacity sc on ad.AIRLINE_ID = SC.AIRLINE_ID
order by seat_utilization desc)
```

```
select
AIRLINE_ID, avg(seat_utilization) as avg_seat_utilization
from seat_utilization
group by 1
order by avg_seat_utilization desc;
```

AIRLINE_ID	avg_seat_utilization
21656	100
19537	100
21610	100
22135	100
22127	100
21022	99.76278334211914
20273	99.16805324459234
21386	97.05882352941177
21614	94.43447037701975

#7. Identify popular destination airport based on inbound passenger counts.

```
select
```

```

DEST_AIRPORT_ID, DEST_CITY_NAME, sum(PASSENGERS) as inbound_passengers
from airport_project_data
group by DEST_AIRPORT_ID, DEST_CITY_NAME
order by inbound_passengers desc
limit 10; #fortop10.

```

DEST_AIRPORT_ID	DEST_CITY_NAME	inbound_passengers
10397	Atlanta, GA	3126527
11292	Denver, CO	2714566
11298	Dallas/Fort Worth, TX	2590043
13930	Chicago, IL	2199886
13204	Orlando, FL	1941202
12889	Las Vegas, NV	1912476
14107	Phoenix, AZ	1820495
11057	Charlotte, NC	1808085
12892	Los Angeles, CA	1800089

#8. Examin the relationship between city poulation & airport passenger traffic.
select*from all_city_pop;

```

with outgoing_passengers as
(select
ORIGIN_AIRPORT_ID, sum(PASSENGERS) as tot_passengers
from airport_project_data
group by 1),

```

```

incoming_passengers as
(select
DEST_AIRPORT_ID, sum(PASSENGERS) as tot_passengers
from airport_project_data
group by 1),

```

```

all_airport as
(select
distinct ORIGIN_AIRPORT_ID as airport_id,
ORIGIN_CITY_NAME as city_name
from airport_project_data
union
select distinct DEST_AIRPORT_ID as airport_id,
DEST_CITY_NAME as city_name
from airport_project_data),

```

```

passenger_traffic AS (
  SELECT
    SUBSTRING_INDEX(aa.city_name, ',', 1) AS city_name,
    COALESCE(op.tot_passengers,0) +

```

```

        COALESCE(ip.tot_passengers,0) AS total_passenger_travelling
FROM all_airport aa
LEFT JOIN outgoing_passengers op
    ON aa.airport_id = op.ORIGIN_AIRPORT_ID
LEFT JOIN incoming_passengers ip
    ON aa.airport_id = ip.DEST_AIRPORT_ID
)

```

```

select cp.city_name, cp.population, pt.total_passenger_travelling
from passenger_traffic pt
join all_city_pop cp
on pt.city_name = cp.city_name
where CP.population is not null
and total_passenger_travelling is not null
and population <>0
order by population asc;

```

city_name	population	total_passenger_travelling
Chadron	5324	28
Bar Harbor	5356	816
Kenai	5399	8735
Manteo	5403	0
Homer	5697	3221
Lewisburg	5712	638
East Farmingdale	6222	467
Ambler	6521	188
Penn Yan	6529	2